Chapter 5 - Antiderivatives

Calculus I

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Section 1

Antiderivatives and Indefinite Integrals

Antiderivatives

Problem: Given a function $f(x)$ does there exist a function $F(x)$ so that

$$F'(x) = f(x)?$$

Definition 5.1.1 (Antiderivative)

Given a function $f(x)$, an antiderivative is a function $F(x)$ such that

$$F'(x) = f(x)$$

If $\boldsymbol{F'(x) = f(x)}$ for all x in an interval I, we say that $\boldsymbol{F(x)}$ is an antiderivative for $f(x)$ on I.

Antiderivatives

Example 5.1.2

Consider $f(x) = x^2$.
Let $\boldsymbol{F}(x) = \frac{x^3}{3}$. Then

$$F'(x) = \frac{3x^{3-1}}{3} = x^2 = f(x)$$

so $\boldsymbol{F}(x) = \frac{x^3}{3}$ is an antiderivative of $f(x)$.

Note: Notice that

$$G(x) = \frac{x^3}{3} + 2$$

is also an antiderivative of $f(x) = x^2$.

Antiderivatives

Recall: If a function $h(x)$ is constant on an open interval $\boldsymbol{I}$, then $h'(x) = 0$ for all $x \in I$.

Important Observation: Given any function $f(x)$, if $F(x)$ is an antiderivative of $f(x)$, then so is

$$G(x) = F(x) + C$$

for any $C \in \mathbb{R}$.

Question: Are all antiderivatives of $f(x)$ of the form

$$G(x) = F(x) + C$$

for some $C \in \mathbb{R}$?

Constant Function Theorem

Theorem 5.1.3 (Constant Function Theorem)

Assume that $f'(x) = 0$ *for all* $x \in I$*, then there exists a* $\alpha \in \mathbb{R}$ *such that* $f(x) = \alpha$ *for every* $x \in I$*.*

Proof: Let x_1 be any point in I and let

$$f(x_1) = \alpha$$

Pick any other $x_2 \in \boldsymbol{I}$. Then the MVT guarantees us that there exists a c between x_1 and x_2 with

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Then since $f'(c) = 0$, we have
and hence

$$0 = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$f(x_2) = f(x_1) = \alpha$$

Constant Function Theorem

Observation: We know that if $f(x) = e^x$, then

$$f(x) = f'(x)$$

for all $x \in \mathbb{R}$.

Question: Are there other functions $g(x)$ with

$$g(x) = g'(x)?$$

Constant Function Theorem

Example 5.1.4

Show that if $g(x)$ is such that

$$g(x) = g'(x)$$

for all $\boldsymbol{x} \in \mathbb{R}$, there exists a $\boldsymbol{C} \in \mathbb{R}$ so that

$$g(x) = Ce^x$$

Constant Function Theorem

Solution: Let

$$h(x) = \frac{g(x)}{e^x}$$

Differentiate $h(x)$ using the quotient rule to get

$$\begin{aligned} h'(x) &= \frac{e^x g'(x) - \frac{d}{dx}\left(e^x\right) g(x)}{\left(e^x\right)^2} \\ &= \frac{e^x g(x) - e^x g(x)}{e^{2x}} \\ &= 0 \end{aligned}$$

since $g'(x) = g(x)$ and $\frac{d}{dx}\left(e^x\right) = e^x$. So there exists $C \in \mathbb{R}$ with

$$h(x) = \frac{g(x)}{e^x} = C \Rightarrow g(x) = Ce^x$$

Antiderivatives

Theorem 5.1.5 (The Antiderivative Theorem)

Assume that $f'(x) = g'(x)$ *for all* $x \in I$. *Then there exists an* α *such that*

$$f(x) = g(x) + \alpha$$

for every $x \in I$.

Proof: Let

$$H(x) = f(x) - g(x)$$

Then

$$H'(x) = f'(x) - g'(x) = 0$$

for each $x \in I$. Therefore, there exists $\alpha \in \mathbb{R}$ so that

$$H(x) = \alpha \Rightarrow f(x) = g(x) + \alpha$$

for all $x \in I$.

Indefinite Integrals

Leibniz Notation for Antiderivatives:

We will denote the family of antiderivatives of a function $f(x)$ by

$$\int f(x)dx$$

For example,

$$\int x^2 dx = \frac{x^3}{3} + C$$

The symbol

$$\int f(x)dx$$

is called the indefinite integral of $f(x)$. The function $f(x)$ is called the integrand.

Indefinite Integrals

Theorem 5.1.6 (Power Rule for Antiderivatives)

If $\alpha \neq -1$, *then*

$$\int x^{\alpha} dx = \frac{x^{\alpha+1}}{\alpha+1} + C$$

Note: To check that this theorem is correct we need only differentiate.

Since

$$\frac{d}{dx}\left(\frac{x^{\alpha+1}}{\alpha+1} + C\right) = x^{\alpha}$$

we have found all of the antiderivatives of x^{α}.

Indefinite Integrals

Examples:

1. $$\int \frac{1}{x} dx = \ln(|x|) + C$$

2. $$\int e^x dx = e^x + C$$

3. $$\int \sin(x) dx = -\cos(x) + C$$

4. $$\int \cos(x) dx = \sin(x) + C$$

5. $$\int \sec^2(x) dx = \tan(x) + C$$

Indefinite Integrals

Examples:

6.
$$\int \frac{1}{1+x^2}dx = \arctan(x) + C$$

7.
$$\int \frac{1}{\sqrt{1-x^2}}dx = \arcsin(x) + C$$

8.
$$\int \frac{-1}{\sqrt{1-x^2}}dx = \arccos(x) + C$$

Remark: It can be shown that there is no nice function that represents

$$\int e^{x^2}dx$$

Section 2

Change of Variables for the Indefinite Integral

Change of Variables for the Indefinite Integral

Key Problem:

Are there methods to find antiderivatives of complicated functions?

Antiderivatives and the Chain Rule

Recall: The Chain Rule tells us that if

$$H(x) = f(g(x))$$

then

$$H'(x) = \frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Question: Is there a method to undo the Chain Rule?

Change of Variables for the Indefinite Integral

Remark: Assume we have two functions h and f with

$$h'(u) = f(u)$$

Then

$$\int f(u)du = h(u) + C$$

Let $u = g(x)$ be a function of x. The Chain Rule says that if

$$\boldsymbol{H}(\boldsymbol{x}) = h(g(x))$$

then

$$\begin{aligned} H'(x) &= h'(g(x))g'(x) \\ &= f(g(x))g'(x) \end{aligned}$$

It follows that

$$\begin{aligned} \int f(g(x))g'(x)dx &= H(x) + C \\ &= h(g(x)) + C \\ &= h(u) + C \\ &= \int f(u)du \end{aligned}$$

Change of Variables for the Indefinite Integral

Definition 5.2.1 (Change of Variables Formula)

Let $\boldsymbol{u = g(x)}$ be differentiable. Then

$$\int f(g(x))g'(x)dx = \int f(u)du$$

Notation: We will use the notation

$$h(u)|_{u=g(x)}$$

to mean replace u by $g(x)$ in the formula for $h(u)$.

As such the symbol

$$\int f(u)du\Big|_{u=g(x)}$$

means replace u by $g(x)$ once the antiderivative has been found.

Method of Substitution

Method of Substitution: Evaluate $\int 2xe^{x^2}dx$.

Solution: Let $u = g(x) = x^2$, then $g'(x) = 2x$. If we also let $f(u) = e^u$, then

$$\int 2xe^{x^2}dx = \int f(g(x))g'(x)dx$$

We get that

$$\begin{aligned}
\int 2xe^{x^2}dx &= \int f(g(x))g'(x)dx \\
&= \int f(u)du\Big|_{u=g(x)} \\
&= \int e^u du\Big|_{u=x^2} \\
&= e^u|_{u=x^2} + C \\
&= e^{x^2} + C
\end{aligned}$$

Method of Substitution

Method of Substitution (continued):

1. Start with

$$\int f(g(x))g'(x)dx$$

2. Identify the possible substitution

$$u = g(x)$$

3. Differentiating both sides gives us

$$\frac{du}{dx} = g'(x)$$

4. If we treat du and dx as if they were "numbers", then

$$du = g'(x)dx$$

Method of Substitution

We can now substitute $\boldsymbol{u}$ for $\boldsymbol{g}(x)$ and du for $g'(x)dx$ to get

$$\int f(g(x))g'(x)dx = \int f(u)du\Big|_{u=g(x)}$$

which is the Change of Variables formula.

Method of Substitution

Example 5.2.2

Evaluate $\int \frac{2x}{1+x^2}dx$ by making the substitution $u = 1 + x^2$.

Method of Substitution

If

$$u = 1 + x^2$$

then

$$du = 2xdx$$

Substituting $u = 1 + x^2$ and $du = 2xdx$ into the original integral gives us

$$\int \frac{2x}{1+x^2}dx = \int \frac{1}{u}du\Big|_{u=1+x^2}$$

but

$$\int \frac{1}{u}du = \ln(|u|) + C$$

Hence

$$\begin{aligned}\int \frac{2x}{1+x^2}dx &= \int \frac{1}{u}du\Big|_{u=1+x^2} \\ &= \ln(|u|)|_{u=1+x^2} + C \\ &= \ln\left(\left|1+x^2\right|\right) + C\end{aligned}$$

Method of Substitution

Example 5.2.3

Evaluate $\int x \cos\left(x^2\right) dx$.

Note: Using the substitution $u = x^2$ causes us to be out by a constant factor. However, this will not be a problem by proceeding as follows.

Let $u = x^2$ and $f(u) = \cos(u)$, then we get

$$du = 2xdx$$

We can rearrange the last equation to get that

$$\frac{1}{2}du = xdx$$

Method of Substitution

Substituting $\frac{1}{2}du$ for xdx and u for x^2 we get

$$\begin{aligned}
\int x\cos\left(x^2\right)dx &= \int \frac{1}{2}\cos(u)du\Big|_{u=x^2} \\
&= \frac{1}{2}\int \cos(u)du\Big|_{u=x^2} \\
&= \frac{1}{2}\sin(u)\Big|_{u=x^2} + C \\
&= \frac{1}{2}\sin\left(x^2\right) + C
\end{aligned}$$

Method of Substitution

Example 5.2.4

Evaluate $\int \sec(\theta)d\theta$.

Remark: This does not appear to be an integral that can be evaluated by using substitution. However, we can make the following observation:

$$\begin{aligned}\sec(\theta) &= \sec(\theta)\left(\frac{\sec(\theta)+\tan(\theta)}{\sec(\theta)+\tan(\theta)}\right)\\ &= \frac{\sec^2(\theta)+\sec(\theta)\tan(\theta)}{\sec(\theta)+\tan(\theta)}\end{aligned}$$

so that

$$\int \sec(\theta)d\theta = \int \frac{\sec^2(\theta)+\sec(\theta)\tan(\theta)}{\sec(\theta)+\tan(\theta)}d\theta$$

Method of Substitution

Example (continued): To evaluate

$$\int \sec(\theta)d\theta = \int \frac{\sec^2(\theta) + \sec(\theta)\tan(\theta)}{\sec(\theta) + \tan(\theta)}d\theta$$

let $u = \sec(\theta) + \tan(\theta)$ so

$$du = \sec^2(\theta) + \sec(\theta)\tan(\theta)d\theta$$

We get

$$\begin{aligned}\int \sec(\theta)d\theta &= \int \frac{\sec^2(\theta) + \sec(\theta)\tan(\theta)}{\sec(\theta) + \tan(\theta)}d\theta \\ &= \int \frac{du}{u} \\ &= \ln(|u|) + C \\ &= \ln(|\sec(\theta) + \tan(\theta)|) + C\end{aligned}$$

Trig Identities

Recall: The following are important trigonometric identities:

1. $\sin^2(\theta) + \cos^2(\theta) = 1$
2. $\tan^2(\theta) + 1 = \sec^2(\theta)$
3. $\cos^2(\theta) = \frac{\cos(2\theta)+1}{2}$

Example

Example 5.2.5

Evaluate $\int \frac{1}{1-x^2}dx$.

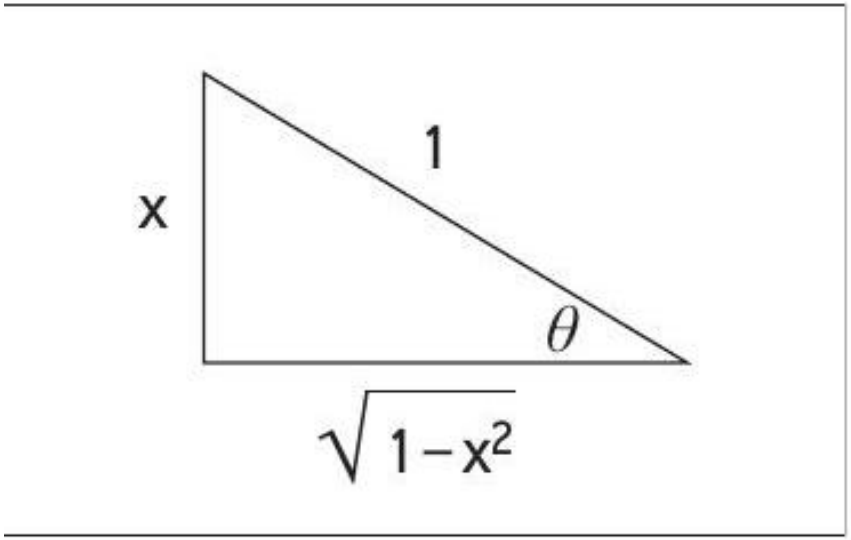
We make the substitution

$$x = \sin(\theta) \Rightarrow \theta = \arcsin(x), dx = \cos(\theta)d\theta$$

$$\begin{aligned}\int \frac{1}{1-x^2}dx &= \int \frac{\cos(\theta)}{1-\sin^2(\theta)}d\theta \\ &= \int \frac{\cos(\theta)}{\cos^2(\theta)}d\theta \\ &= \int \sec(\theta)d\theta \\ &= \ln|\sec(\theta)+\tan(\theta)| + C\end{aligned}$$

Problem: How do we convert this back to a function of x ?

Example



Note: We have $x = \sin(\theta) \Rightarrow \theta = \arcsin(x), -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

$$\Rightarrow \quad \cos(\theta) = \sqrt{1-x^2}$$

$$\Rightarrow \quad \sec(\theta) = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \quad \tan(\theta) = \frac{x}{\sqrt{1-x^2}}$$

Therefore

$$\int \frac{1}{1-x^2}\,dx = \ln\left|\sec(\theta)+\tan(\theta)\right| + C = \ln\left|\frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}\right| + C$$

Example

Example 5.2.6

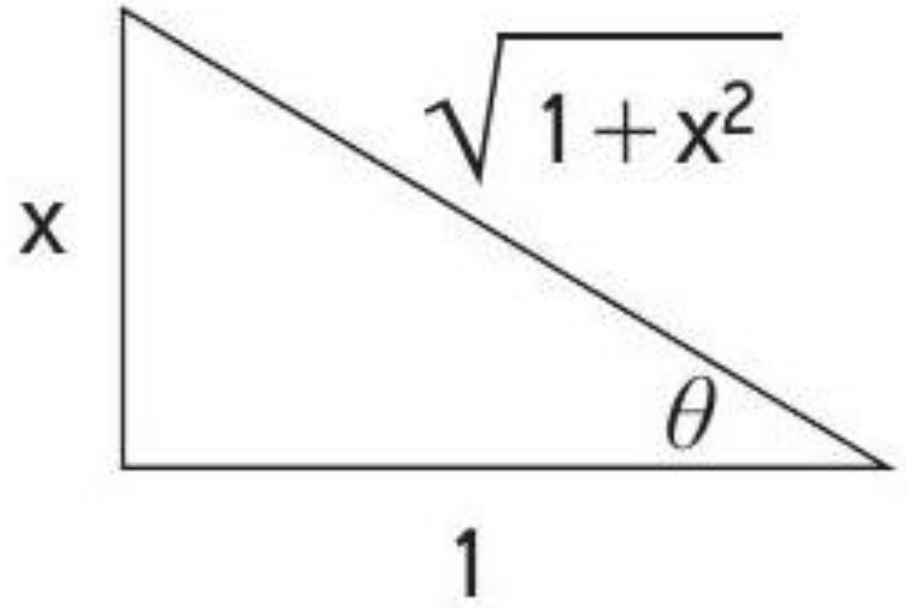
Evaluate $\int \frac{1}{(x^2+1)^2}\,dx$.

Example

We make the substitution $\theta = \arctan(x) \Rightarrow x = \tan(\theta), dx = \sec^2(\theta)d\theta$.

$$\begin{aligned}
\int \frac{1}{\left(x^2+1\right)^2}dx &= \int \frac{\sec^2(\theta)}{\left(\tan^2(\theta)+1\right)^2}d\theta \\
&= \int \frac{\sec^2(\theta)}{\left(\sec^2(\theta)\right)^2}d\theta \\
&= \int \frac{1}{\sec^2(\theta)}d\theta \\
&= \int \cos^2(\theta)d\theta \\
&= \int \left(\frac{\cos(2\theta)}{2}+\frac{1}{2}\right)d\theta \\
&= \frac{1}{2}\cdot\frac{\sin(2\theta)}{2}+\frac{\theta}{2}+C \\
&= \frac{1}{4}(\sin(2\theta))+\frac{\theta}{2}+C
\end{aligned}$$

Example

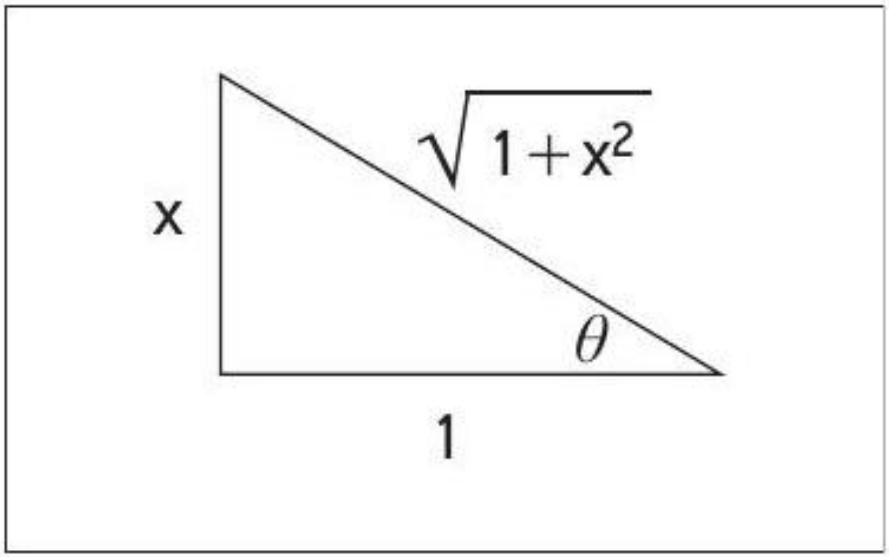


Note: We have $\theta = \arctan(x) \Rightarrow x = \tan(\theta), -\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

$$\Rightarrow \quad \cos(\theta) = \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow \quad \sin(\theta) = \frac{x}{\sqrt{1+x^2}}$$

Example



Therefore

$$\int \frac{1}{(x^2+1)^2}dx = \frac{1}{4}(\sin(2\theta)) + \frac{\theta}{2} + C = \frac{1}{2}(\sin(\theta)\cos(\theta)) + \frac{\theta}{2} + C$$

$$= \frac{1}{2}\left(\frac{x}{1+x^2} + \arctan(x)\right) + C$$

Summary

Expression	Substitution
$a^2 - x^2$	$x = a\sin(\theta), -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$
$a^2 + x^2$	$x = a\tan(\theta), -\frac{\pi}{2} < \theta < \frac{\pi}{2}$
$x^2 - a^2$	$x = a\sec(\theta), 0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$

Section 3

Integration by Parts

The Integration by Parts Formula

Recall: The Product Rule for derivatives states that

$$\begin{aligned}\frac{d}{dx}(f \cdot g)(x) &= \left(\frac{df}{dx}\right)(x) \cdot g(x) + f(x) \cdot \left(\frac{dg}{dx}\right)(x) \\ &= f'(x)g(x) + f(x)g'(x)\end{aligned}$$

Key Observation: To undo the Product Rule we get that if f and g are differentiable,

$$f(x)g(x) = \int \frac{d}{dx}(f(x)g(x))dx = \int f'(x)g(x)dx + \int f(x)g'(x)dx$$

Definition 5.3.1 (Integration by Parts Formula)

$$\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$$

or

$$\int f'(x)g(x)dx = f(x)g(x) - \int f(x)g'(x)dx$$

The Strategy

Example 5.3.2

Evaluate $\int xe^x dx$.

Strategy:

- The goal is to rid the integrand of the x.
- The method to remove the $\boldsymbol{x}$ is to differentiate it "downwards" to produce the constant 1 .
- As a compensation, we must integrate the other factor e^x "upwards".
- Integrating e^x is not difficult.

The Strategy

Example 5.3.2 (continued)

Evaluate $\int xe^x dx$.

Solution: Let

$$f(x) = x \text{ and } g'(x) = e^x$$

Then

$$f'(x) = 1 \text{ and } g(x) = e^x$$

Integrating by parts, we have

$$\begin{aligned}\int xe^x dx = \int f(x)g'(x)dx &= f(x)g(x) - \int f'(x)g(x)dx \\ &= xe^x - \int 1 \cdot e^x dx \\ &= xe^x - e^x + C\end{aligned}$$

More Examples

Example 5.3.3

Evaluate $\int x^2 \sin(x)dx$.

Strategy: We would like to eliminate the x^2 term. We can do this by differentiating twice $\Rightarrow$ using Integration by Parts twice!

Step 1: Let $f(x) = x^2, g'(x) = \sin(x)$. Then $f'(x) = 2x$ and $g(x) = -\cos(x)$.

$$\begin{aligned}\int x^2 \sin(x)dx &= -x^2 \cos(x) - \int 2x(-\cos(x))dx \\ &= -x^2 \cos(x) + 2\int x\cos(x)dx\end{aligned}$$

More Examples

Example 5.3.3 (continued)

Evaluate $\int x^2 \sin(x)dx$.

Step 2: Evaluate $\int x\cos(x)dx$.

Let $f(x) = x, g'(x) = \cos(x)$. Then $f'(x) = 1$ and $g(x) = \sin(x)$.

$$\begin{aligned}\int x\cos(x)dx &= x\sin(x) - \int \sin(x)dx \\ &= x\sin(x) + \cos(x) + C\end{aligned}$$

Therefore

$$\begin{aligned}\int x^2\sin(x)dx &= -x^2\cos(x) + 2\int x\cos(x)dx \\ &= -x^2\cos(x) + 2x\sin(x) + 2\cos(x) + C\end{aligned}$$

More Examples

Example 5.3.4

Evaluate $\int e^x \sin(x)dx$.

Strategy: Nothing can be eliminated by differentiation but we can take advantage of the cyclic nature of $\sin(x)$ and $\cos(x)$ with respect to differentiation.

Let $f(x) = \sin(x), g'(x) = e^x$. Then $f'(x) = \cos(x)$ and $g(x) = e^x$ gives

$$\int e^x \sin(x)dx = e^x \sin(x) - \int e^x \cos(x)dx$$

Apply Integration by Parts again with $f(x) = \cos(x)$ and $g'(x) = e^x$ to get

$$\begin{aligned}\int e^x \sin(x)dx &= e^x \sin(x) - \int e^x \cos(x)dx \\ &= e^x \sin(x) - \left[e^x \cos(x) + \int e^x \sin(x)dx\right] \\ &= e^x \sin(x) - e^x \cos(x) - \int e^x \sin(x)dx\end{aligned}$$

More Examples

Example 5.3.4 (continued)

Evaluate $\int e^x \sin(x) dx$.

Hence

$$2 \int e^x \sin(x) dx = e^x \sin(x) - e^x \cos(x)$$

and

$$\int e^x \sin(x) dx = \frac{e^x \sin(x) - e^x \cos(x)}{2} + C$$

More Examples

Example 5.3.5

Evaluate $\int \ln(x)dx = \int 1 \cdot \ln(x)dx$

Note: There is an "invisible" second factor in the integrand: $1 \cdot \ln(x)$

Let $g'(x) = 1$ and $f(x) = \ln(x)$. Then $f'(x) = \frac{1}{x}$ and $g(x) = x$

$$\begin{aligned}\int \ln(x)dx = \int f(x)g'(x)dx &= f(x)g(x) - \int f'(x)g(x)dx \\ &= x\ln(x) - \int \frac{1}{x} \cdot xdx \\ &= x\ln(x) - \int 1dx \\ &= x\ln(x) - x + C\end{aligned}$$

More Examples

Example 5.3.6

Evaluate $\int \arctan(x)dx = \int 1 \cdot \arctan(x)dx$

Let $f(x) = \arctan(x)$ and $g'(x) = 1$. Then $f'(x) = \frac{1}{1+x^2}$ and $g(x) = x$. We obtain

$$\begin{aligned}\int \arctan(x)dx = \int f(x)g'(x)dx &= f(x)g(x) - \int f'(x)g(x)dx \\ &= x\arctan(x) - \int \frac{x}{1+x^2}dx\end{aligned}$$

More Examples

Example 5.3.6 (continued)

Evaluate $\int \arctan(x)dx = \int 1 \cdot \arctan(x)dx$

Making the substitution $u = 1 + x^2 \Rightarrow du = 2xdx$ we now have

$$\begin{aligned}
\int \frac{x}{1+x^2}dx &= \frac{1}{2}\int \frac{2x}{1+x^2}dx \\
&= \frac{1}{2}\int \frac{1}{u}du \\
&= \frac{1}{2}\ln(|u|) + C_0 \\
&= \frac{1}{2}\ln\left(\left|1+x^2\right|\right) + C_0
\end{aligned}$$

More Examples

Example 5.3.6 (continued)

Evaluate $\int \arctan(x)dx = \int 1 \cdot \arctan(x)dx$

Therefore

$$\begin{aligned}\int \arctan(x)dx &= x\arctan(x) - \int \frac{x}{1+x^2}dx \\ &= x\arctan(x) - \left(\frac{1}{2}\ln\left(\left|1+x^2\right|\right) + C_0\right) \\ &= x\arctan(x) - \frac{1}{2}\ln\left(\left|1+x^2\right|\right) + C\end{aligned}$$

Section 4

Partial Fractions

Partial Fractions

Recall: A function of the form

$$f(x) = \frac{p(x)}{q(x)}$$

where $\boldsymbol{p}$ and $\boldsymbol{q}$ are polynomials is called a rational function.

Problem: Evaluate

$$\int f(x)dx = \int \frac{p(x)}{q(x)}dx$$

Example

Example: Evaluate $\int \frac{1}{x^2-1}dx$.

Strategy: Look for constants $\boldsymbol{A}$ and $\boldsymbol{B}$ so that

$$\frac{1}{x^2-1} = \frac{1}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1}$$

then

$$\begin{aligned}
\int \frac{1}{x^2-1}dx &= \int \frac{1}{(x+1)(x-1)}dx \\
&= \int \left(\frac{A}{x+1} + \frac{B}{x-1}\right)dx \\
&= \int \frac{A}{x+1}dx + \int \frac{B}{x-1}dx \\
&= A\ln(|x+1|) + B\ln(|x-1|) + C
\end{aligned}$$

Example

Example (continued): Evaluate $\int \frac{1}{x^2-1} dx$..

Question: How do we find $\boldsymbol{A}$ and $\boldsymbol{B}$?

Observation: We have

$$\frac{1}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1} = \frac{A(x-1)+B(x+1)}{(x+1)(x-1)}$$

so

$$1 = A(x-1) + B(x+1)$$

Let

$$x = 1 \Rightarrow 1 = A(1-1) + B(1+1) \Rightarrow 1 = 2B \Rightarrow B = \frac{1}{2}$$

Let

$$x = -1 \Rightarrow 1 = A(-1-1) + B(-1+1) \Rightarrow 1 = -2A \Rightarrow A = -\frac{1}{2}$$

Example

Example (continued): Evaluate $\int \frac{1}{x^2-1}dx$

Therefore

$$\frac{1}{(x+1)(x-1)} = \frac{-\frac{1}{2}}{x+1} + \frac{\frac{1}{2}}{x-1}$$

and

$$\int \frac{1}{x^2-1}dx = -\frac{1}{2}\ln(|x+1|) + \frac{1}{2}\ln(|x-1|) + C$$

Type I Partial Fraction Decomposition

Definition 5.4.1 (Type I Partial Fraction Decomposition)

Assume that

$$f(x) = \frac{p(x)}{q(x)}$$

where p and q are polynomials such that

1. $\text{degree}(p(x)) < \text{degree}(q(x)) = k$,
2. $q(x)$ can be factored into the product of k linear terms each with distinct roots. That is

$$q(x) = a\,(x - a_1)\,(x - a_2)\,(x - a_3)\cdots(x - a_k)$$

where the a_i's are all different and none of the a_i's are roots of $p(x)$.
Then there exists constants $\boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{A}_3, \cdots, \boldsymbol{A}_k$ such that

$$f(x) = \frac{1}{a}\left[\frac{A_1}{x - a_1} + \frac{A_2}{x - a_2} + \frac{A_3}{x - a_3} + \cdots + \frac{A_k}{x - a_k}\right]$$

We say that f admits a Type I partial fraction decomposition.

Type I Partial Fraction Decomposition

Theorem 5.4.2 (Integration of Type I Partial Fractions)

Assume that $f(x) = \frac{p(x)}{q(x)}$ *admits a Type I partial fraction decomposition of the form*

$$f(x) = \frac{1}{a}\left[\frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \cdots + \frac{A_k}{x-a_k}\right]$$

Then

$$\begin{aligned}\int f(x)dx =& \frac{1}{a}\left[\int \frac{A_1}{x-a_1}dx + \int \frac{A_2}{x-a_2}dx + \cdots\right.\\ &\left. + \int \frac{A_k}{x-a_k}dx\right]\\ =& \frac{1}{a}\left[A_1 \ln\left(|x-a_1|\right) + A_2 \ln\left(|x-a_2|\right) + \cdots\right.\\ &\left. + A_k \ln\left(|x-a_k|\right)\right] + C\end{aligned}$$

Type I Partial Fraction Decomposition

Example: We know that

$$\frac{x}{(x+2)(x-1)(x-2)} = \frac{A_1}{x+2} + \frac{A_2}{x-1} + \frac{A_3}{x-2}$$

So

$$x = A_1(x-1)(x-2) + A_2(x+2)(x-2) + A_3(x+2)(x-1)$$

Substituting for the roots $x = -2, x = 1$ and $x = 2$ gives

$$A_1 = \frac{-2}{(-2-1)(-2-2)} = -\frac{1}{6}, \quad (x = -2)$$

$$A_2 = \frac{1}{(1+2)(1-2)} = -\frac{1}{3}, \quad (x = 1)$$

and

$$A_3 = \frac{2}{(2+2)(2-1)} = \frac{1}{2}, \quad (x = 2)$$

Hence $\int \frac{x}{(x+2)(x-1)(x-2)}dx = -\frac{1}{6}\ln(|x+2|) - \frac{1}{3}\ln(|x-1|) + \frac{1}{2}\ln(|x-2|) + C$

Type I Partial Fraction Decomposition

Key Observation:

If

$$\frac{p(x)}{q(x)} = \frac{1}{a}\left[\frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \frac{A_3}{x-a_3} + \cdots + \frac{A_k}{x-a_k}\right]$$

then

$$A_m = \frac{a \cdot p\left(a_m\right)}{\prod_{\substack{j=1 \\ j \neq m}}^{k}\left(a_m - a_j\right)}$$

where

$$\prod_{j=1}^{k} b_j = b_1 \cdot b_2 \cdot b_3 \cdots b_k$$

Type II Partial Fraction Decomposition

Definition 5.4.3 (Type II Partial Fraction Decomposition)

Assume that

$$f(x) = \frac{p(x)}{q(x)}$$

where $\boldsymbol{p(x)}$ and $\boldsymbol{q(x)}$ are polynomials such that

1. ddegree$(p(x)) <$ degree$(q(x)) = k$,
2. $q(x)$ can be factored into the product of linear terms with non-distinct roots

$$q(x) = a\,(x - a_1)^{m_1}\,(x - a_2)^{m_2}\,(x - a_3)^{m_3} \cdots (x - a_l)^{m_l}$$

where at least one of the m_j's is greater than 1.

Type II Partial Fraction Decomposition

Definition 5.4.3 (Type II Partial Fraction Decomposition) (continued)

We say that f admits a Type II partial fraction decomposition as follows: Each expression $(x-a_j)^{m_j}$ in the factorization of $q(x)$ will contribute m_j terms to the decomposition, one for each power of $x-a_j$ from 1 to m_j, which when combined will be of the form

$$\frac{p(x)}{q(x)} = \sum_{j=1}^{l} \frac{A_{j,1}}{x-a_j} + \frac{A_{j,2}}{(x-a_j)^2} + \frac{A_{j,3}}{(x-a_j)^3} + \cdots + \frac{A_{j,m_j}}{(x-a_j)^{m_j}}$$

The number m_j is called the multiplicity of the root a_j.

Type II Partial Fraction Decomposition

Observation: The procedure for integrating a rational function with a Type II partial fraction decomposition is very similar to the procedure for integrating those of Type I.

Example: Evaluate $\int \frac{1}{x^2(x+1)}dx$.

We have

$$\frac{1}{x^2(x+1)} = \frac{A_{1,1}}{x} + \frac{A_{1,2}}{x^2} + \frac{A_{2,1}}{x+1}$$

$$\begin{aligned}\int \frac{1}{x^2(x+1)}dx &= \int \frac{A_{1,1}}{x}dx + \int \frac{A_{1,2}}{x^2}dx + \int \frac{A_{2,1}}{x+1}dx \\ &= A_{1,1}\ln(|x|) - \frac{A_{1,2}}{x} + A_{2,1}\ln(|x+1|) + C\end{aligned}$$

Problem: How do we find $A_{1,1}, A_{1,2}$ and $A_{2,1}$?

Type II Partial Fraction Decomposition

Example (continued): Evaluate $\int \frac{1}{x^2(x+1)}dx$

If

$$\frac{1}{x^2(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1}$$

then multiplying by $q(x)$ gives:

$$1 = Ax(x+1) + B(x+1) + Cx^2$$

Letting $\boldsymbol{x = 0}$ gives

$$1 = B(0+1) \Rightarrow B = 1$$

Letting $x = -1$ gives

$$1 = C(-1)^2 \Rightarrow C = 1$$

Question: How do we find A ?

Type II Partial Fraction Decomposition

Example (continued): Evaluate $\int \frac{1}{x^2(x+1)}dx$.

Observation: We find $\boldsymbol{A}$ by comparing coefficients in the equation

$$1 = Ax(x+1) + B(x+1) + Cx^2$$

We have

$$1 = Ax^2 + Ax + Bx + B + Cx^2 = (A+1)x^2 + (A+1)x + 1$$

$$\begin{aligned} \Rightarrow \quad & 0x^2 = (A+1)x^2 \\ \Rightarrow \quad & 0 = A+1 \\ \Rightarrow \quad & A = -1 \end{aligned}$$

Hence

$$\int \frac{1}{x^2(x+1)}dx = -\ln(|x|) - \frac{1}{x} + \ln(|x+1|) + C$$

Type II Partial Fraction Decomposition

Remark: Not all polynomials factor over the Real numbers into products of linear terms. For example, the polynomial $x^2 + 1$ cannot be factored any further.

This is an example of an irreducible quadratic. In fact, a quadratic $ax^2 + bx + c$ is irreducible if its discriminant

$$b^2 - 4ac < 0$$

Fact: The Fundamental Theorem of Algebra shows that every polynomial $q(x)$ factors in the form

$$q(x) = a\left(p_1(x)\right)^{m_1}\left(p_2(x)\right)^{m_2}\left(p_3(x)\right)^{m_3}\cdots\left(p_k(x)\right)^{m_k}$$

where each $p_i(x)$ is either of the form $(x - a)$ or it is an irreducible quadratic of the form $x^2 + bx + c$.

Type III Partial Fraction Decomposition

Definition 5.4.4 (Type III Partial Fraction Decomposition)

Let $f(x) = \frac{p(x)}{q(x)}$ be a rational function with $\text{degree}(p(x)) < \text{degree}(q(x))$, but $q(x)$ does not factor into linear terms.

We say that f admits a Type III partial fraction decomposition as follows:

Suppose that $q(x)$ has an irreducible factor $x^2 + bx + c$ as above with multiplicity $\boldsymbol{m}$. Then this factor will contribute terms of the form

$$\frac{B_1 x + C_1}{x^2 + bx + c} + \frac{B_2 x + C_2}{(x^2 + bx + c)^2} + \cdots + \frac{B_m x + C_m}{(x^2 + bx + c)^m}$$

to the decomposition.

The linear terms are handled exactly as they were in the previous cases.

Type III Partial Fraction Decomposition

Example: Evaluate $\int \frac{1}{x^3+x} dx$.

Solution: First observe that

$$f(x) = \frac{1}{x^3 + x} = \frac{1}{x\left(x^2 + 1\right)}$$

so there will be constants A, B and C such that

$$f(x) = \frac{1}{x^3 + x} = \frac{1}{x\left(x^2 + 1\right)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

Type III Partial Fraction Decomposition

Example (continued): Evaluate $\int \frac{1}{x^3+x}dx$.

Step 1: To find the constants, we begin by cross multiplying to obtain

$$1 = A\left(x^2+1\right) + (Bx+C)x$$

Step 2: Substitute $x = 0$, the only Real root, to find the coefficient $\boldsymbol{A}$.

This gives

$$1 = A\left(0^2+1\right) + (B(0)+C)(0)$$

or

$$A = 1$$

Type III Partial Fraction Decomposition

Example (continued): Evaluate $\int \frac{1}{x^3+x}dx$.

Step 3: The remaining constants are found by comparing coefficients.

Expanding equation $(*)$ gives:

$$1 = (A+B)x^2 + Cx + A$$

Comparing the coefficients of x^2 gives

$$0 = A + B \Rightarrow B = -A$$

Since $A = 1$, we get

$$B = -1$$

Comparing the coefficients of x gives

$$C = 0$$

Therefore,

$$\frac{1}{x\left(x^2+1\right)} = \frac{1}{x} - \frac{x}{x^2+1}$$

Type III Partial Fraction Decomposition

Example (continued): Evaluate $\int \frac{1}{x^3+x}dx$.

Step 4: We have

$$\begin{aligned}\int \frac{1}{x^3+x}dx &= \int \frac{1}{x\left(x^2+1\right)}dx \\ &= \int \frac{1}{x}dx - \int \frac{x}{x^2+1}dx \\ &= \ln(|x|) - \int \frac{x}{x^2+1}dx\end{aligned}$$

Type III Partial Fraction Decomposition

Example (continued): Evaluate $\int \frac{1}{x^3+x} dx$.

To finish the calculation, we use the substitution $u = x^2 + 1$, so $du = 2x dx$ and $dx = \frac{du}{2x}$ to get

$$\begin{aligned}
\int \frac{x}{x^2+1} dx &= \int \frac{x}{u} \frac{du}{2x} \\
&= \frac{1}{2} \int \frac{1}{u} du \\
&= \frac{1}{2} \ln(|u|) + c \\
&= \frac{1}{2} \ln\left(x^2+1\right) + c
\end{aligned}$$

Putting this all together gives

$$\int \frac{1}{x^3+x} dx = \ln(|x|) - \frac{1}{2} \ln\left(x^2+1\right) + c$$

Type III Partial Fraction Decomposition

Remarks:

1. Because $C = 0$ in the previous example,

$$f(x) = \frac{1}{x^3 + x} = \frac{1}{x\left(x^2 + 1\right)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

and the integral was straightforward. Unfortunately, this is not the case for most Type III partial fractions.

2. All of the partial fraction decompositions required the rational function to satisfy the condition degree $(p(x)) < \text{degree}(q(x))$. If this is not the case, then we need to use long division of polynomials to find polynomials $r(x)$ and $p_1(x)$ such that

$$\frac{p(x)}{q(x)} = r(x) + \frac{p_1(x)}{q(x)}$$

and $\text{degree}\left(p_1(x)\right) < \text{degree}(q(x))$.